

Prajjwal Singh

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Education

Graphic Era Deemed To Be University, Dehradun, Uttarakhand
B.Tech in Computer Science Engineering — Specialization: Cybersecurity

2024 – 2028

Technical Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, HTML5, CSS3
Web & Backend: React.js, Node.js, Express.js, Flask, REST API Development, TailwindCSS, Vite
Security: JWT Authentication, bcrypt Encryption, Rate Limiting, CORS, OWASP, Penetration Testing, Network Security, Vulnerability Assessment, Secure SDLC, IAM, Role-Based Access Control (RBAC)
ML & AI: TensorFlow, Keras, Scikit-learn, LangChain, Deep Learning, Neural Networks, RAG, Semantic Search, FAISS (Vector Databases)
Computer Vision & NLP: OpenCV, MediaPipe, Facial Recognition, Image Processing, NLP Pipelines
Databases: MySQL, MongoDB, SQLite, Firebase, Turso LibSQL, Query Optimization
Cloud & DevOps: AWS (EC2, S3, Lambda), Serverless Architecture, Cloud Deployment, CI/CD
Tools: Git, GitHub, VS Code, Postman, Jupyter Notebook, Linux/Unix, Wireshark

Projects

NovaPass — Secure Password Management & Authentication Platform

[Source Code](#)

- Built a full-stack **password management platform** with encrypted credential storage, implementing **JWT-based session management**, bcrypt hashing, protected API routes, and **Role-Based Access Control (RBAC)** to enforce least-privilege security principles
- Engineered secure **RESTful APIs** and backend services for credential lifecycle management, database interaction, and user-session handling; improved API efficiency and platform modularity following **OWASP secure coding guidelines**
- Integrated responsive frontend (HTML/CSS/JS) with secure backend workflows enabling cross-platform credential management, reducing unauthorized access risk through multi-layer **authentication and authorization** controls
- Tech Stack:** Node.js, Express.js, MongoDB, SQL, JWT, bcrypt, REST APIs, JavaScript, HTML5, CSS3, RBAC, Security Engineering

FriendBook — Dual Interface Social Networking Platform

[Source Code](#)

- Developed a full-stack social platform with a **C-based CLI** and a real-time web interface sharing a cloud database (**Turso/LibSQL + SQLite**), demonstrating **system design** and multi-interface architecture skills
- Implemented **graph-based friend-suggestion engine** (BFS, adjacency list), JWT authentication, friend-request workflows, and admin dashboard; secured with bcrypt, Helmet middleware, and **rate limiting** to prevent abuse
- Tech Stack:** C, Node.js, Express.js, SQLite, Turso LibSQL, JWT, bcrypt, REST APIs, JavaScript, HTML5, CSS3, Graph Algorithms, Data Structures

NewsScope — AI-Powered News Aggregation & Analysis Platform

[Source Code](#)

- Built an **intelligent news aggregation platform** with automated data pipelines, real-time multi-source news retrieval, and **AI-driven content categorization** using scalable backend microservices architecture
- Engineered secure **RESTful APIs** with optimized database queries for dynamic article management, large-scale data processing, and real-time updates; integrated **NLP-based filtering** for category navigation and content discovery
- Tech Stack:** Node.js, Express.js, REST APIs, JavaScript, HTML5, CSS3, Database Management, API Integration, NLP, Data Processing, Backend Engineering

Achievements & Leadership

- State-Level Volleyball Champion** — Demonstrated teamwork, strategic leadership, and high-pressure performance at competitive state level
- Progressive student leadership: **Vice-Captain (2020–21)** → **Captain (2021–22)** → **School President (2022–23)** → **Student Counselor (2023–24)**; mentored peers, facilitated conflict resolution, and drove academic initiatives
- Finalist — SquareX Coding Competition**; demonstrated strong problem-solving, algorithm design, and time-efficient coding under competitive conditions
- 3× Champion — Codex** (Interschool Competitive Programming); consistently ranked top in algorithms, data structures, and optimized code performance